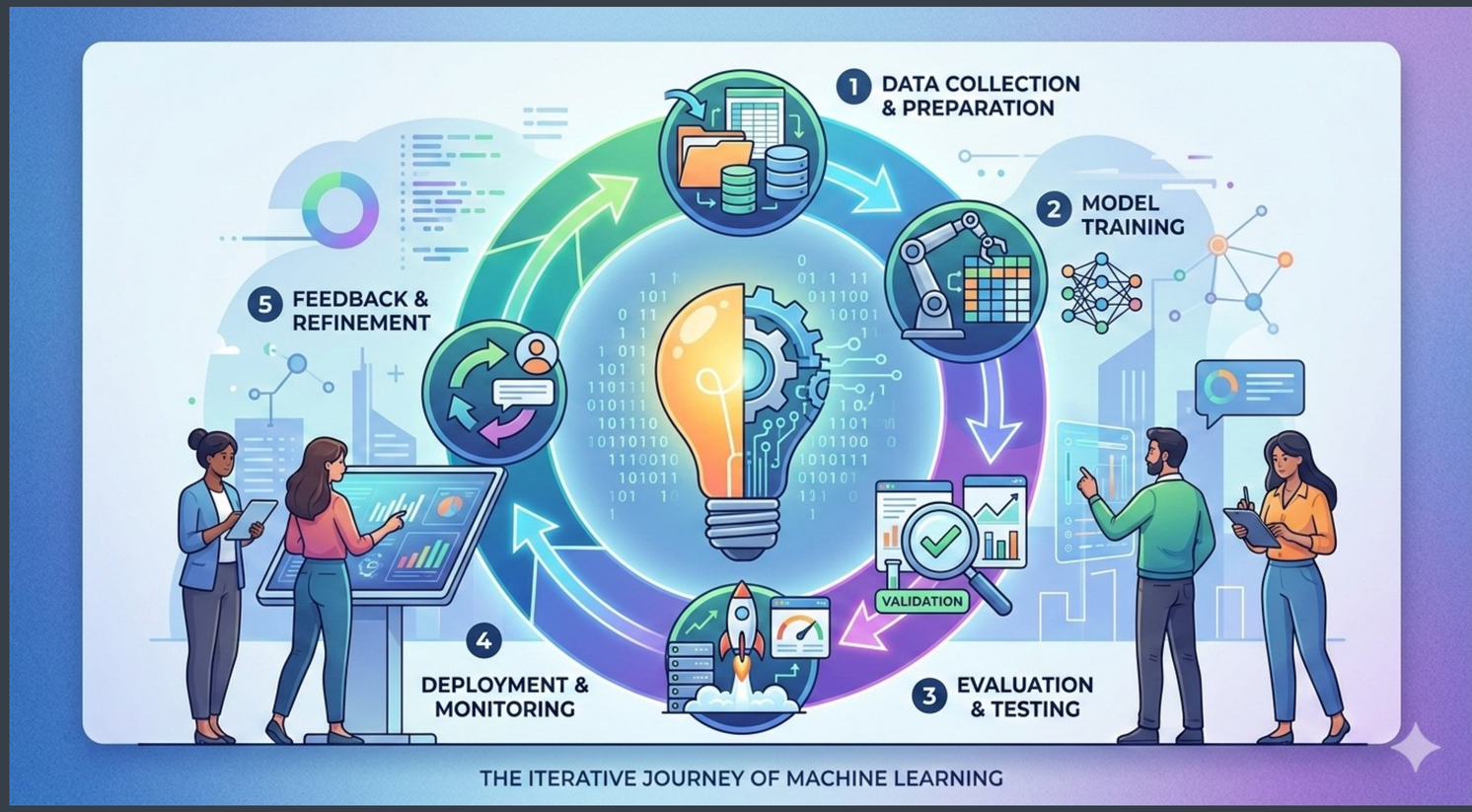
Course Intro

The primary focus of this course is to move beyond 'model training' and address the challenges of collecting, cleaning, and representing data under various assumptions.

Bora GÜNGÖREN

Machine Learning

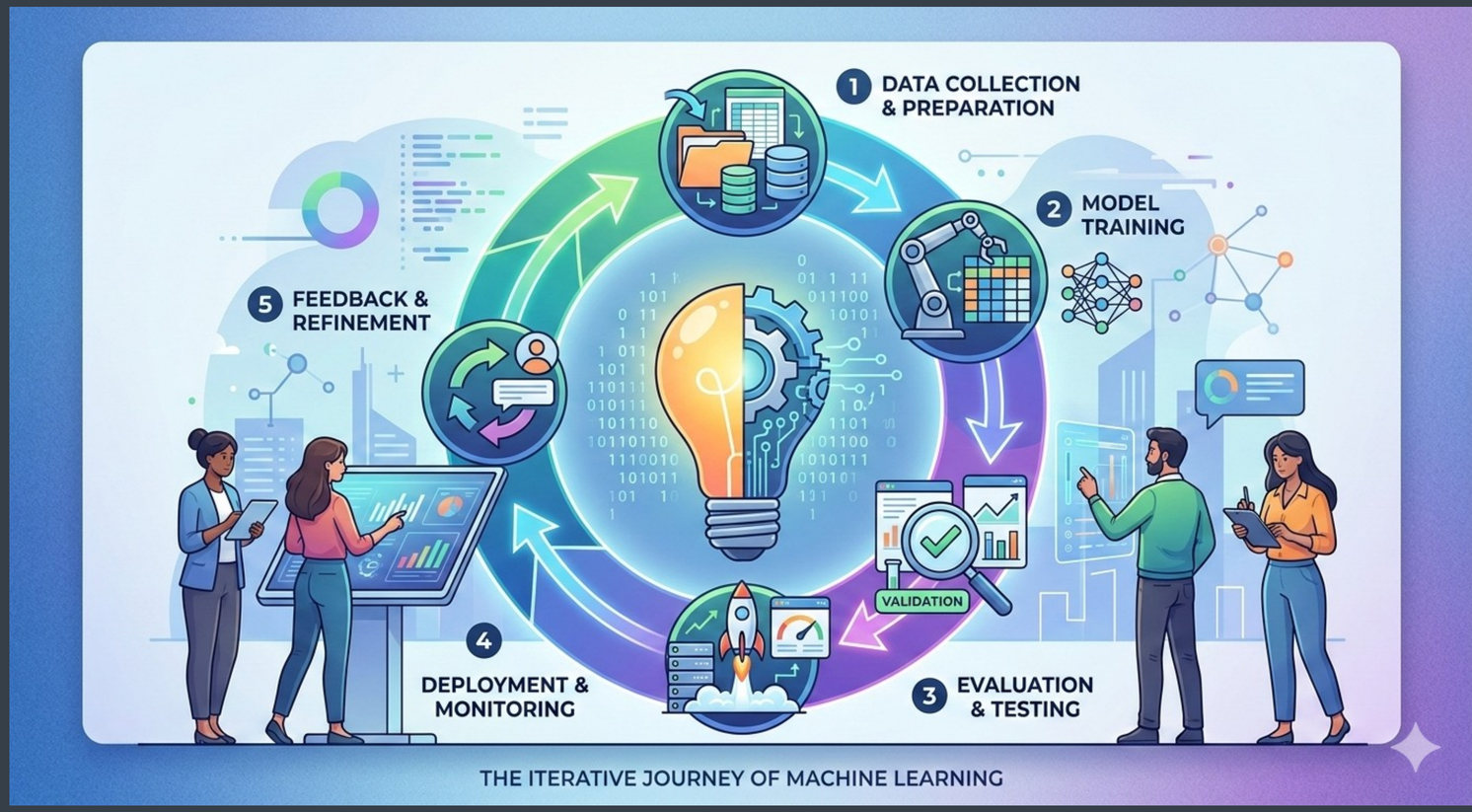
Machine learning is treated not merely as a collection of algorithms, but as an iterative process of data-driven decision-making.



Then what is ML?

Machine Learning

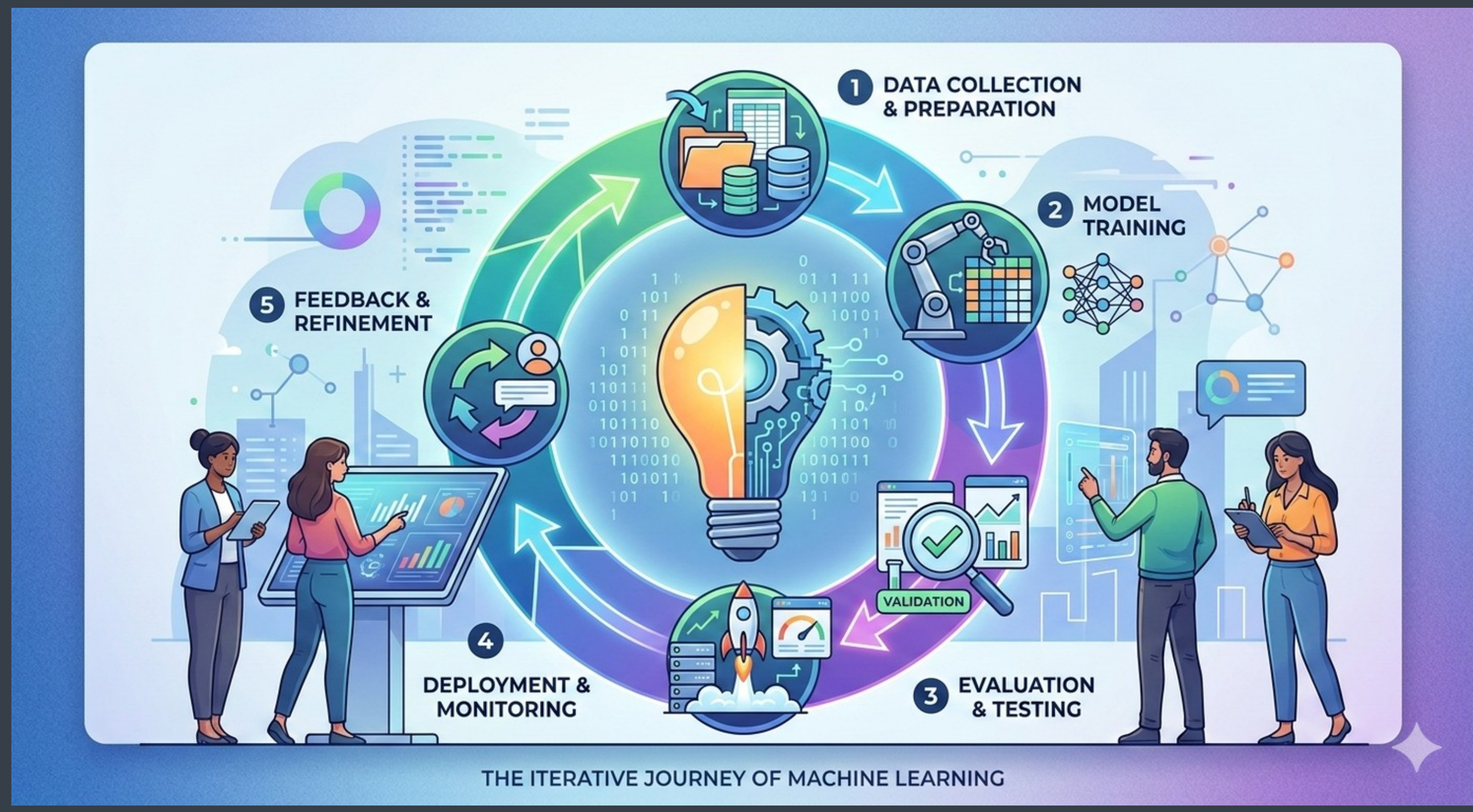
Computer software include algorithms and data structures, both of which are either designed from scratch for a particular very well defined problem, or selected among alternatives based on fitness to a more or less well defined problem.



Machine Learning

What happens when we cannot well define our problem?

When this happens within a tiny scope, data obtained from observations on human (or systems') behavior can be used to estimate and/or imitate what would be done by human (or systemic) actors.



Machine Learning

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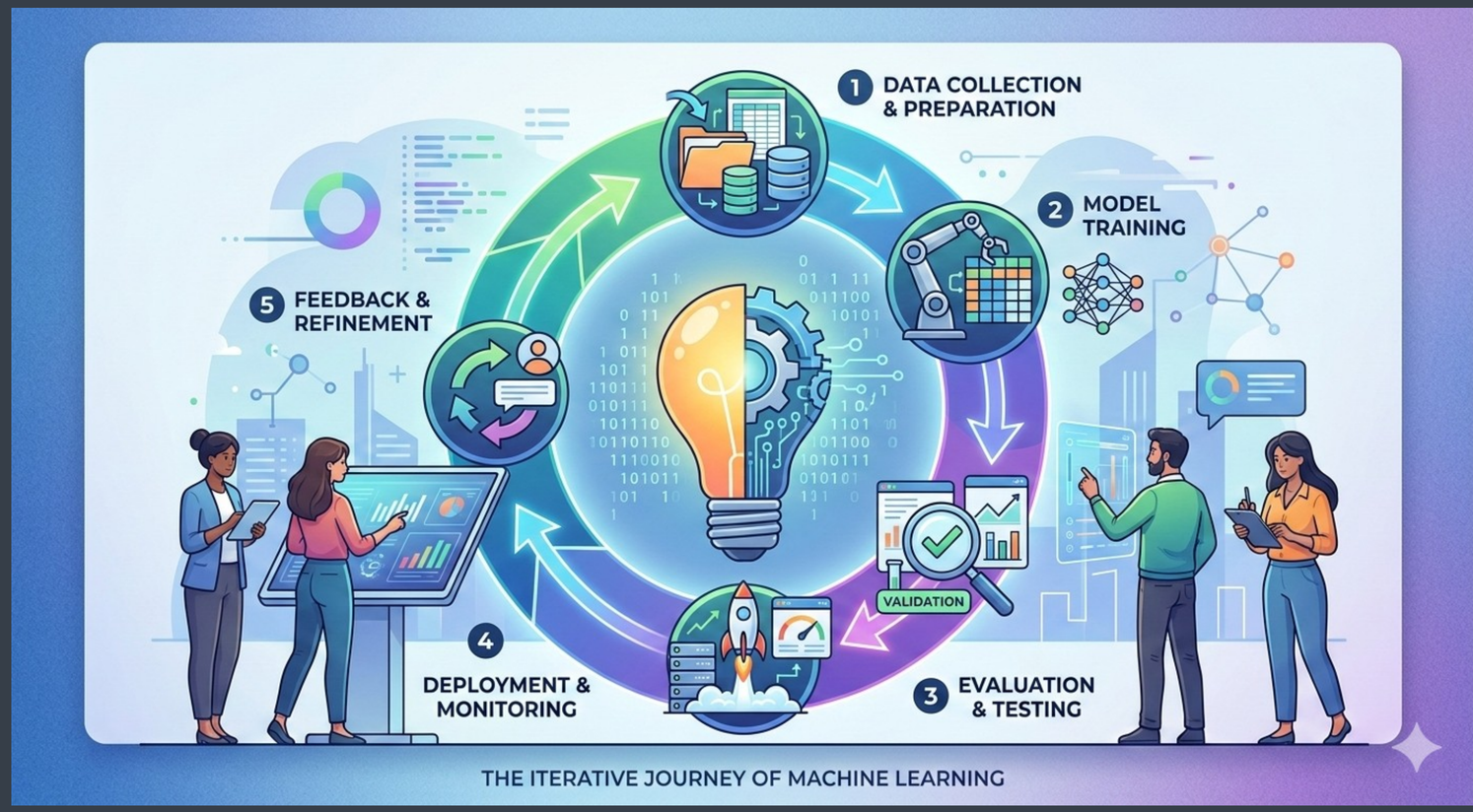
This is called machine learning, the **computer algorithm** learns and mimics existing behavior.

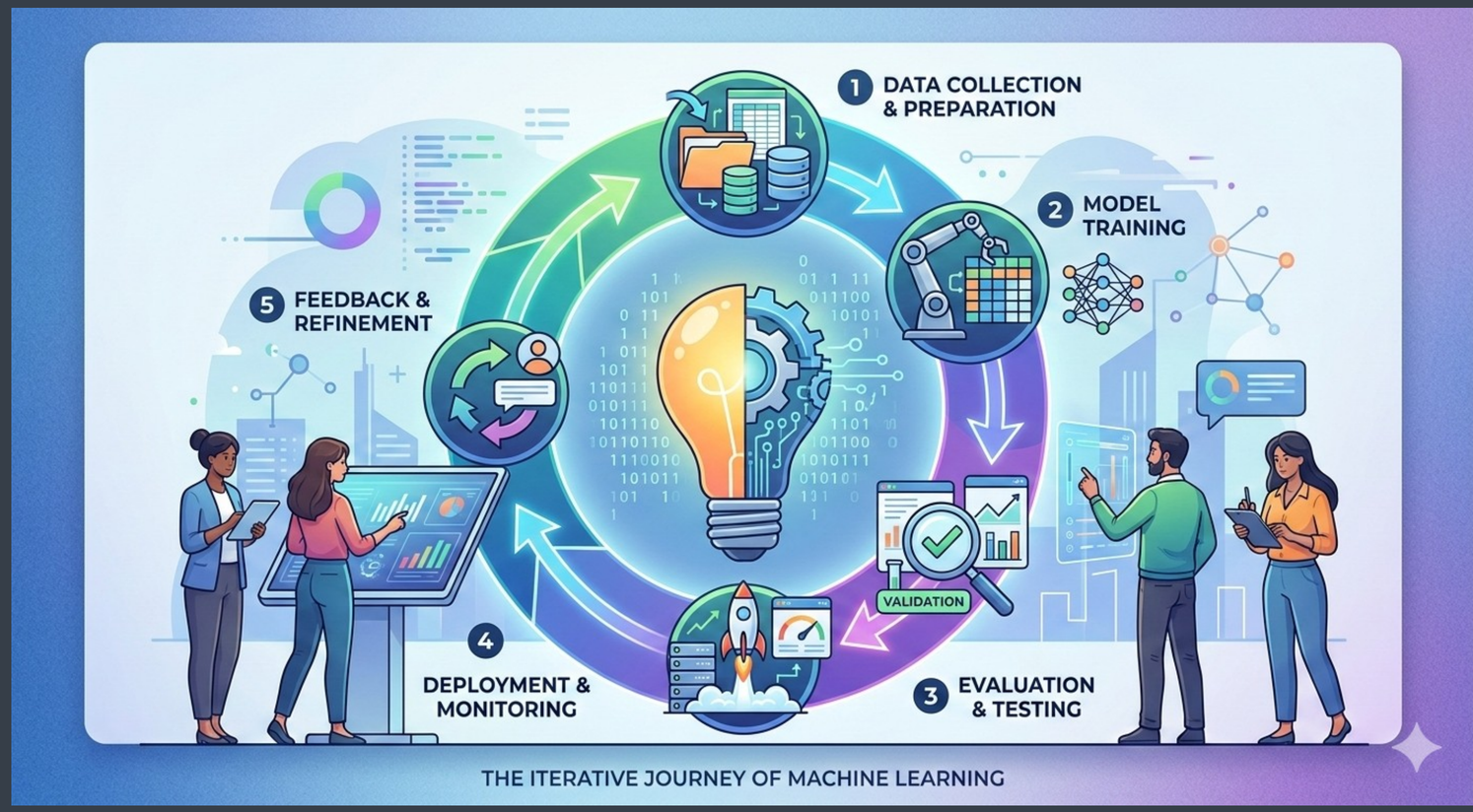


Machine Learning

While a machine learning algorithm is designed to learn and mimic existing behaviors within a defined scope, the algorithm itself is static.

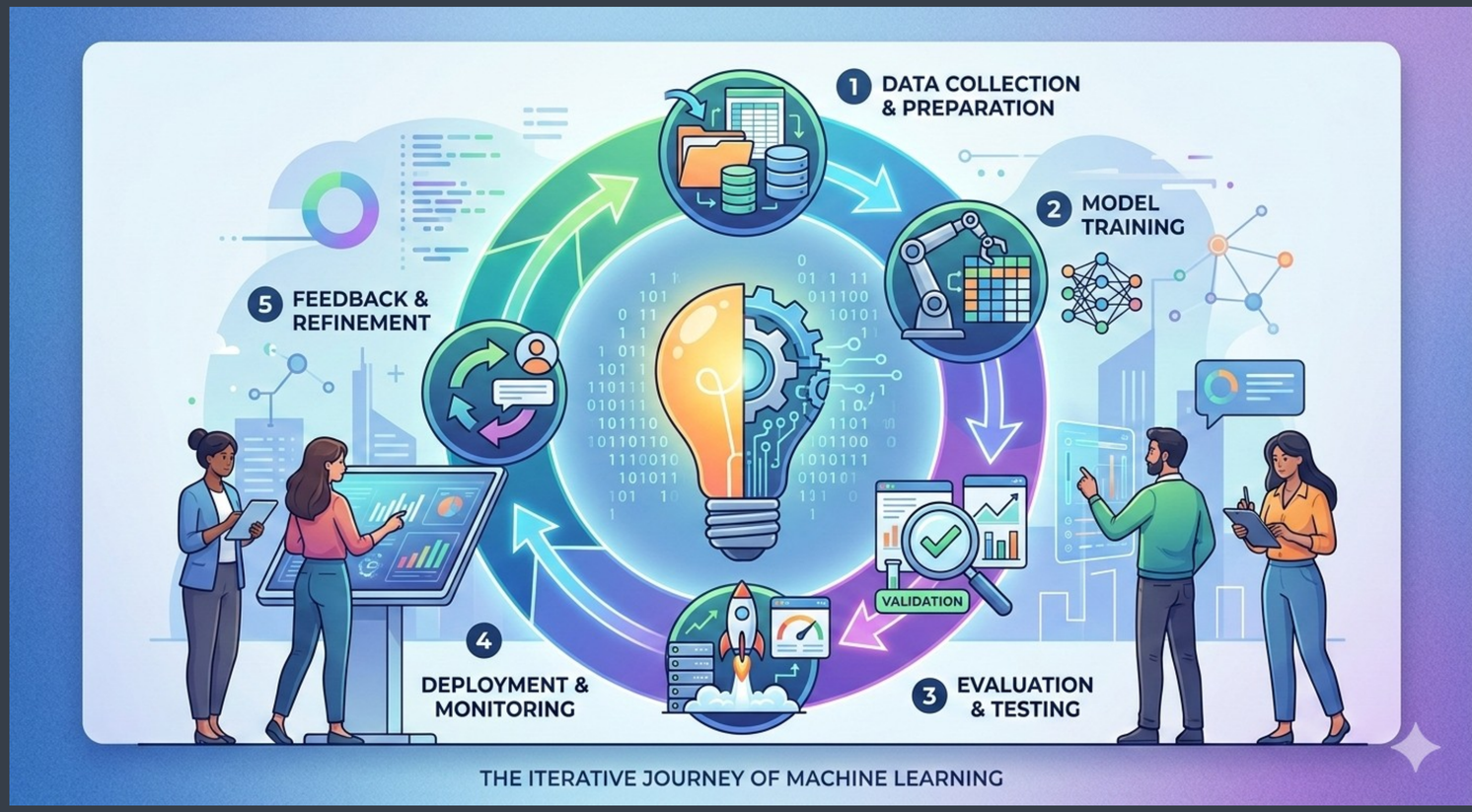
In an academic and professional context, relying solely on the mathematical model creates a "brittle" system.

To move from simple imitation to robust utility, the algorithm must be embedded in a rigorous, iterative process.



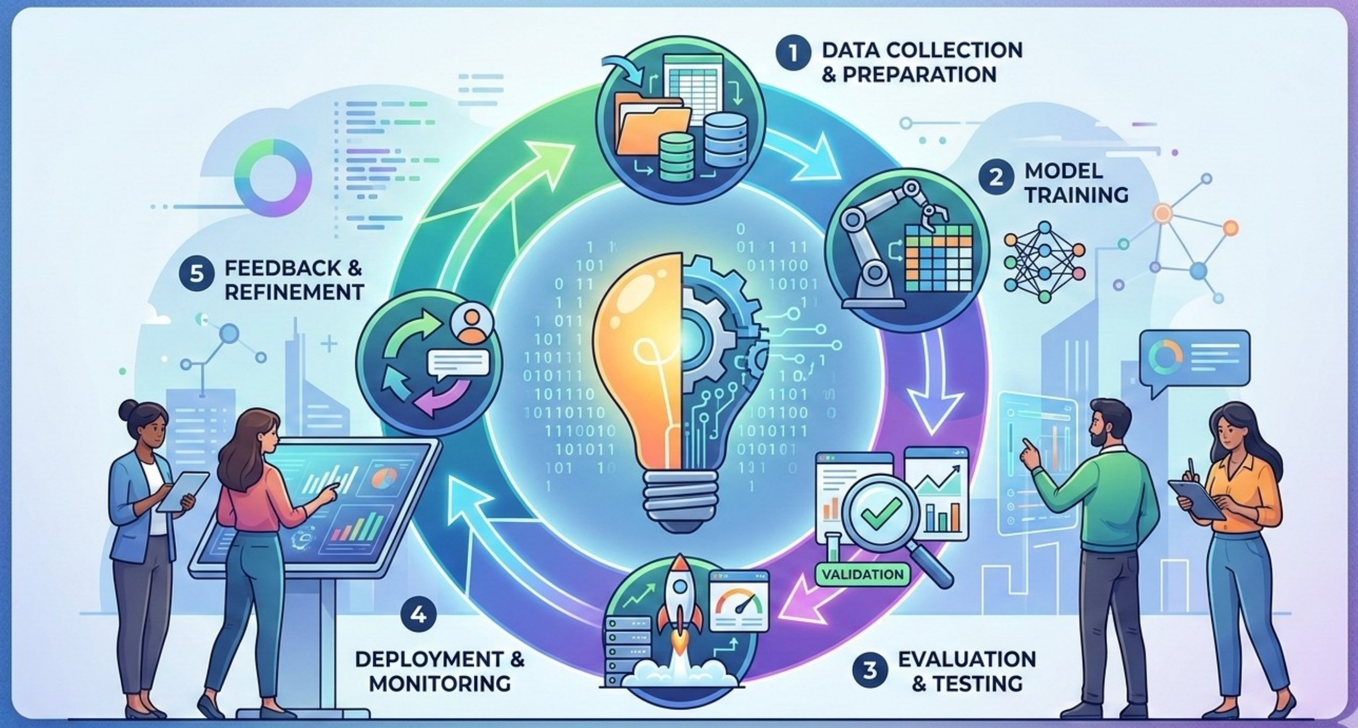
Machine Learning

The efficiency of an algorithm is fundamentally capped by the quality of its input.
- Data as Evidence: In the absence of a well-defined problem, observations are our only ground truth.
 - The Process Role: A structured process ensures data integrity, normalization, and bias mitigation. Without this, the algorithm does not "learn" the problem; it merely codifies the noise and errors present in the sample.



Machine Learning

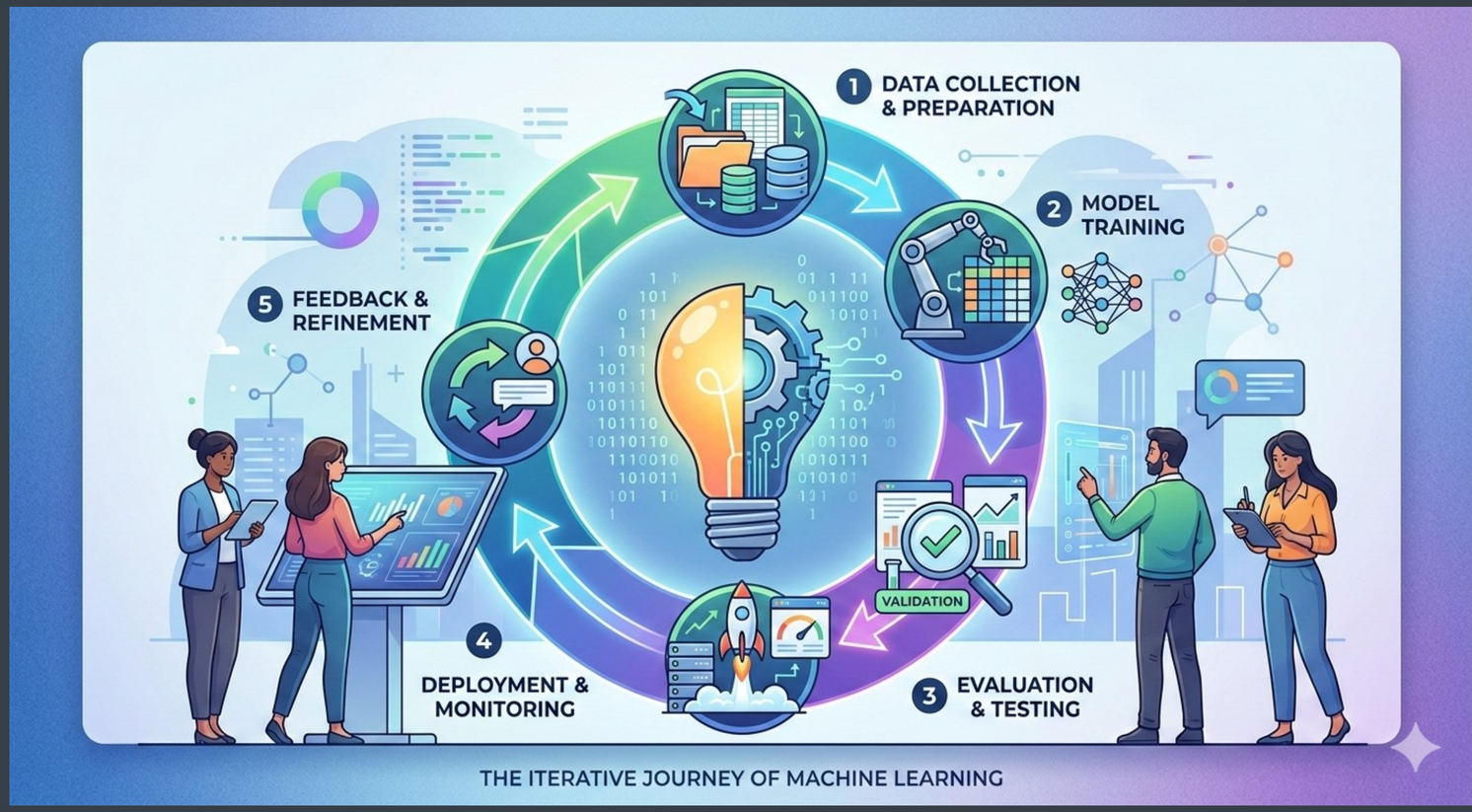
An algorithm can "mimic" behavior perfectly on a training set while failing entirely in the real world (Overfitting).
- Iterative Validation: The process introduces systematic testing against unseen data.
- Efficiency Metric: True algorithmic efficiency is defined by Generalization Error. A process-driven approach ensures the model remains performant across diverse, non-static environments, preventing the "technical debt" of frequent failure.



Machine Learning

Machine learning is not a "set and forget" solution. Because the systems being modeled (human or systemic behavior) are dynamic, the model is subject to Concept Drift.

- Closed-Loop Systems: An effective process incorporates a continuous feedback loop that monitors performance and triggers re-training.
- Strategic Outcome: This transforms ML from a one-time prediction tool into a sustainable decision-support system.



Function

Interaction

Reliability

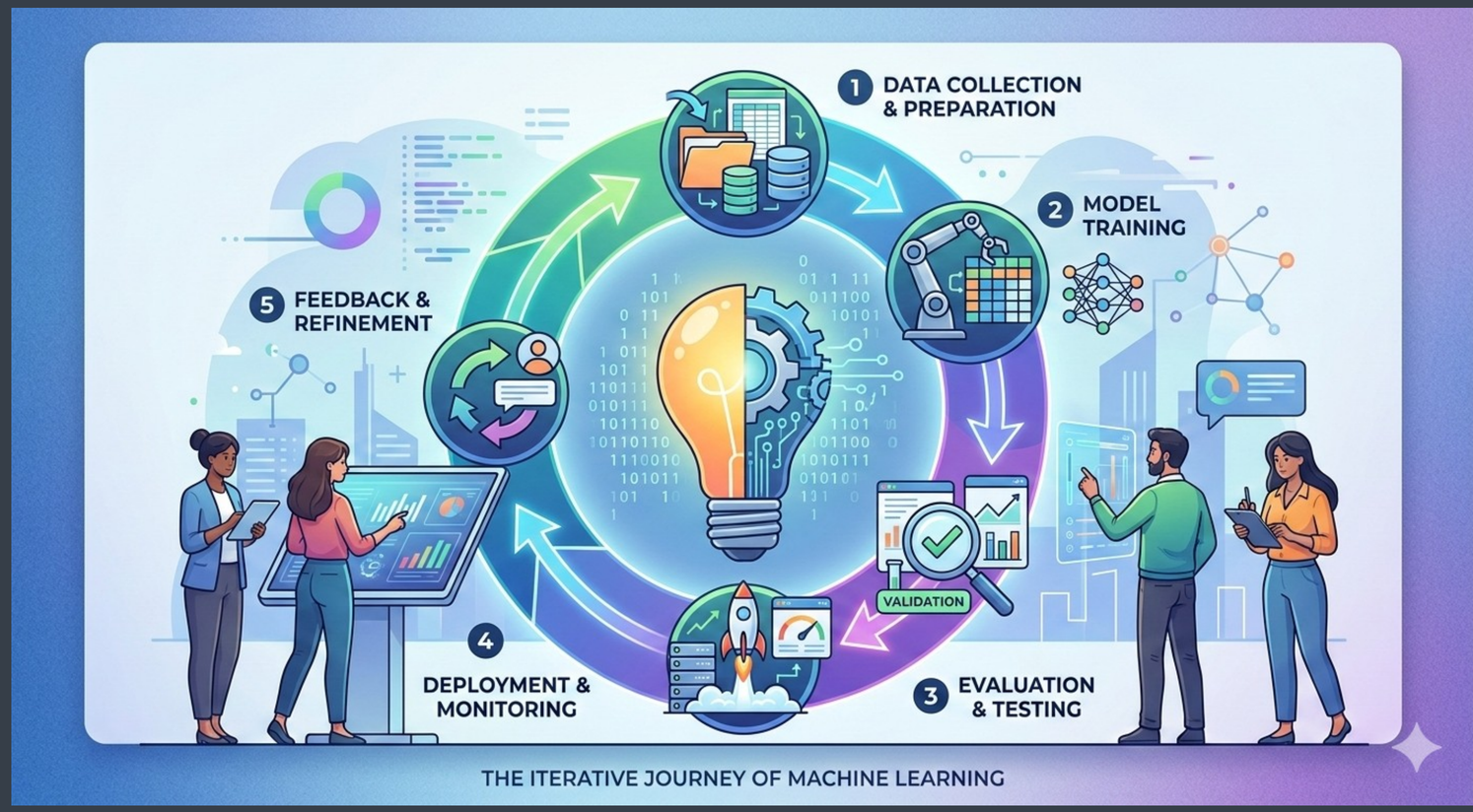
Objective

Dimension	Algorithmic Component (Tactical)	Machine Learning Process (Strategic)
Primary Function	Pattern recognition and mimicry.	Systemic optimization and risk management.
Data Interaction	Processing of a static input vector.	Continuous ingestion, cleaning, and auditing.
Reliability	High variance; prone to bias/overfit.	Controlled through validation and monitoring.
Objective	Minimizing a specific loss function.	Maximizing long-term decision accuracy.

Machine Learning

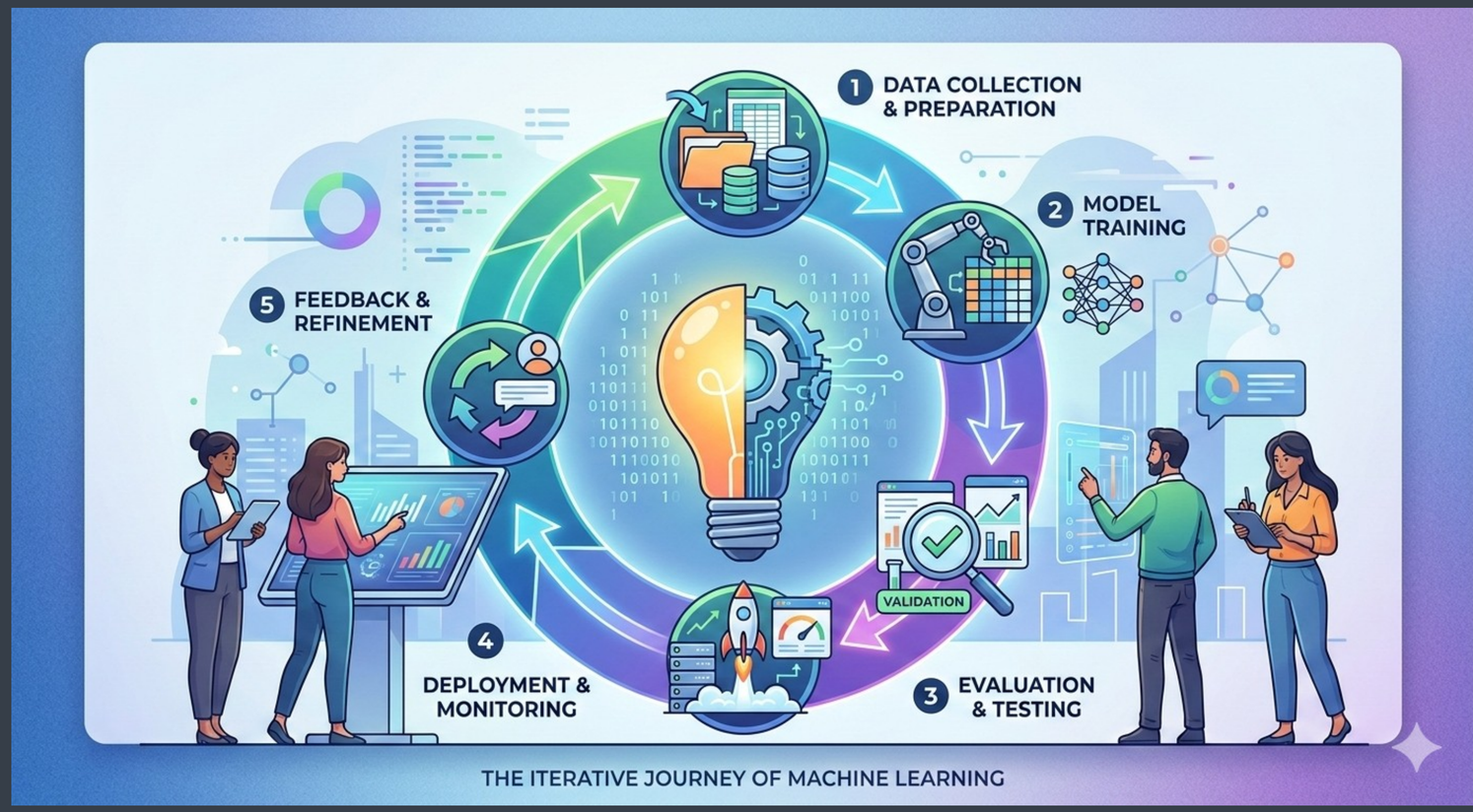
The transition from a raw algorithm to a strategic process is best understood through the CRISP-DM (Cross-Industry Standard Process for Data Mining) or MLOps frameworks.

These frameworks ensure that the "mimicry" described earlier remains aligned with the actual problem being solved.



Why This Course is Different

Last semester, we began with the mathematical foundations of algorithms like polynomial curve fitting. This term, we flip the script. We start with the problem and the data. You won't just study how an algorithm works in a vacuum; you will discover which algorithm you need by building projects.



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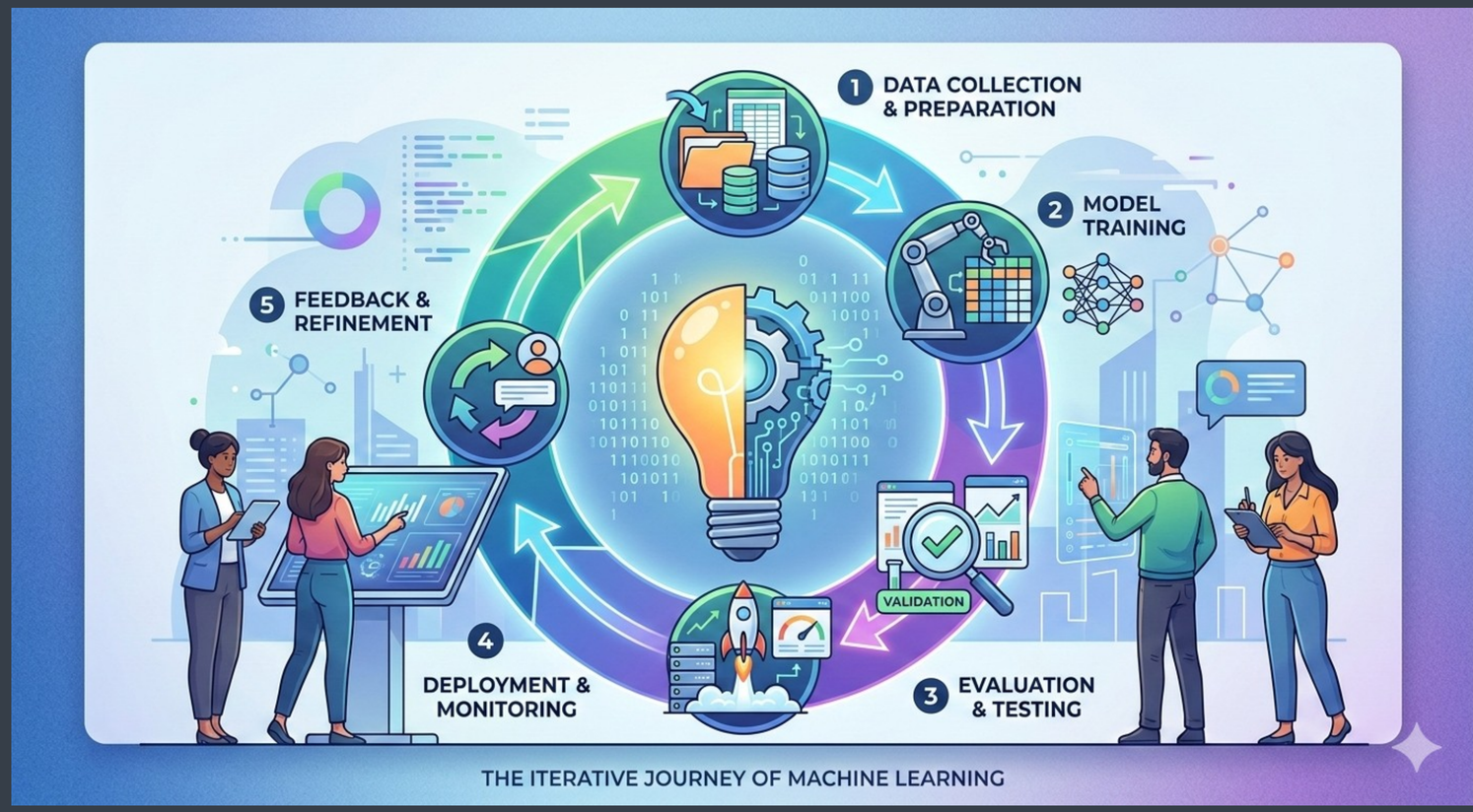
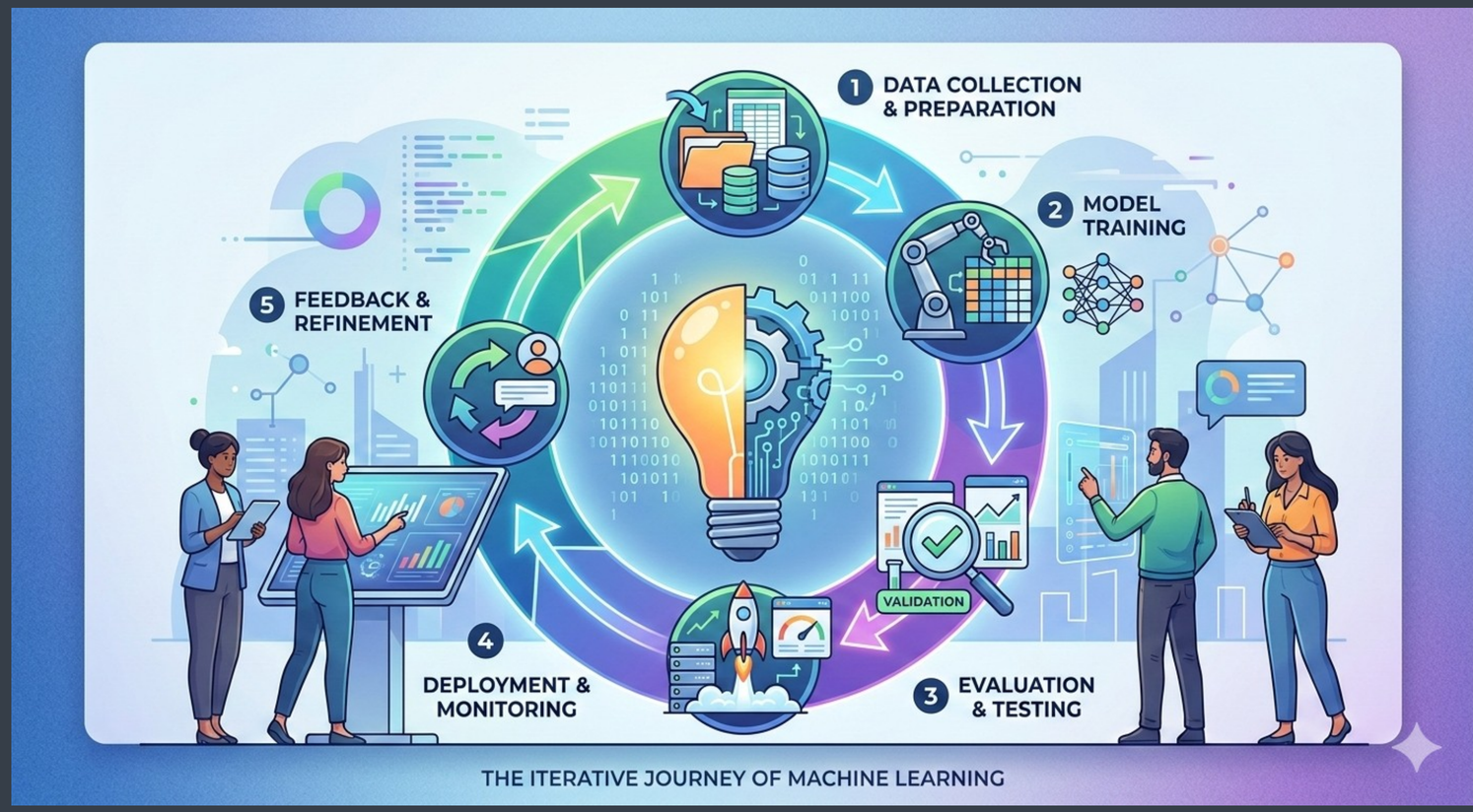


The "AI-Assisted" Policy

We embrace the future of work. Using LLMs like ChatGPT or Gemini for coding is not just allowed—it is encouraged.

However, your value lies in "Analytic Strategy." You must document your prompts and defend every decision.

The AI writes the code, but you must own the logic and the results.

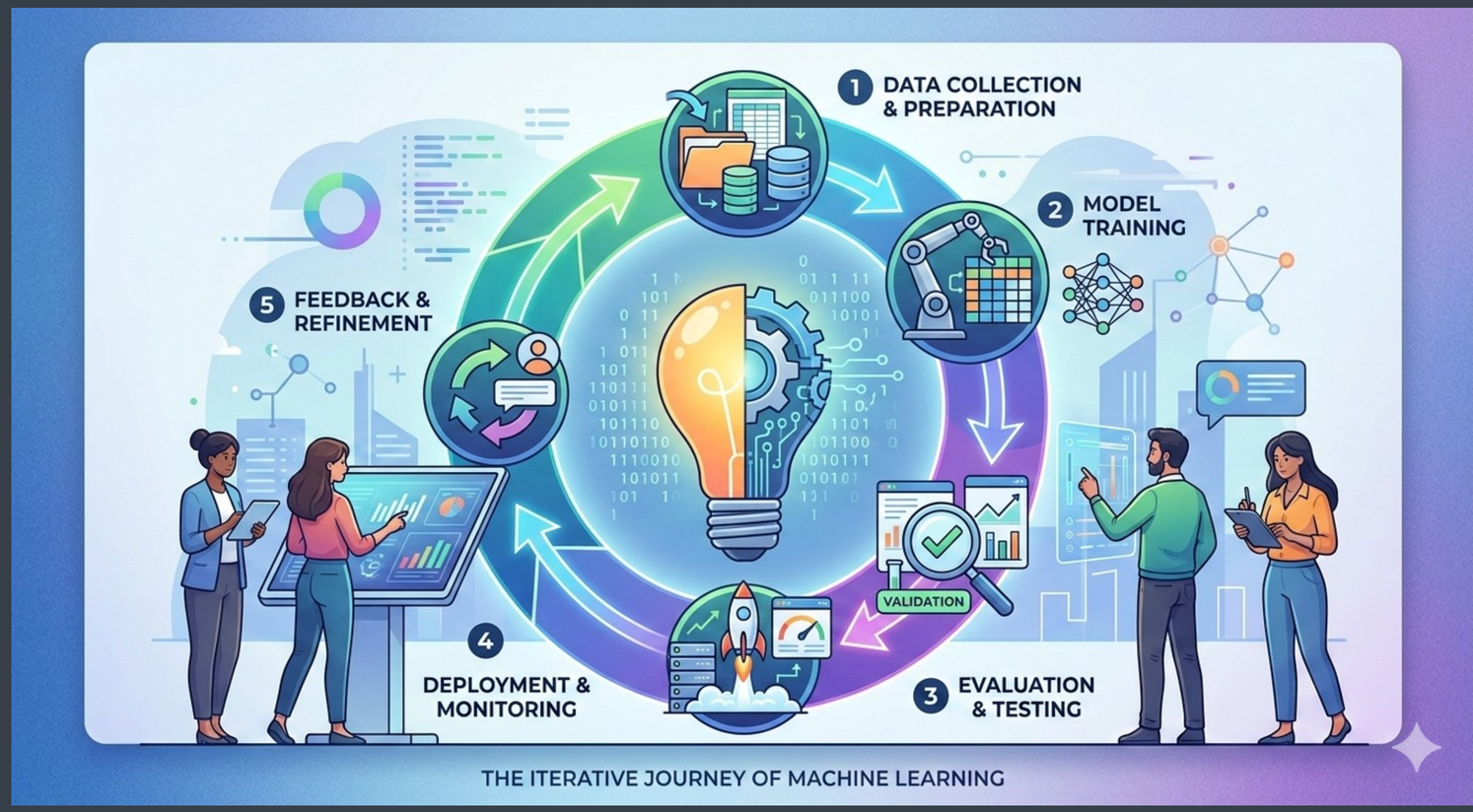


Your New Laboratory

Instead of focusing on R^2 and sum of squared errors in isolation, we focus on the end-to-end pipeline.

You will learn to handle "messy" real-world data, perform feature engineering, and deploy models.

We move beyond academic toy datasets to build solutions that could actually function in a business environment.

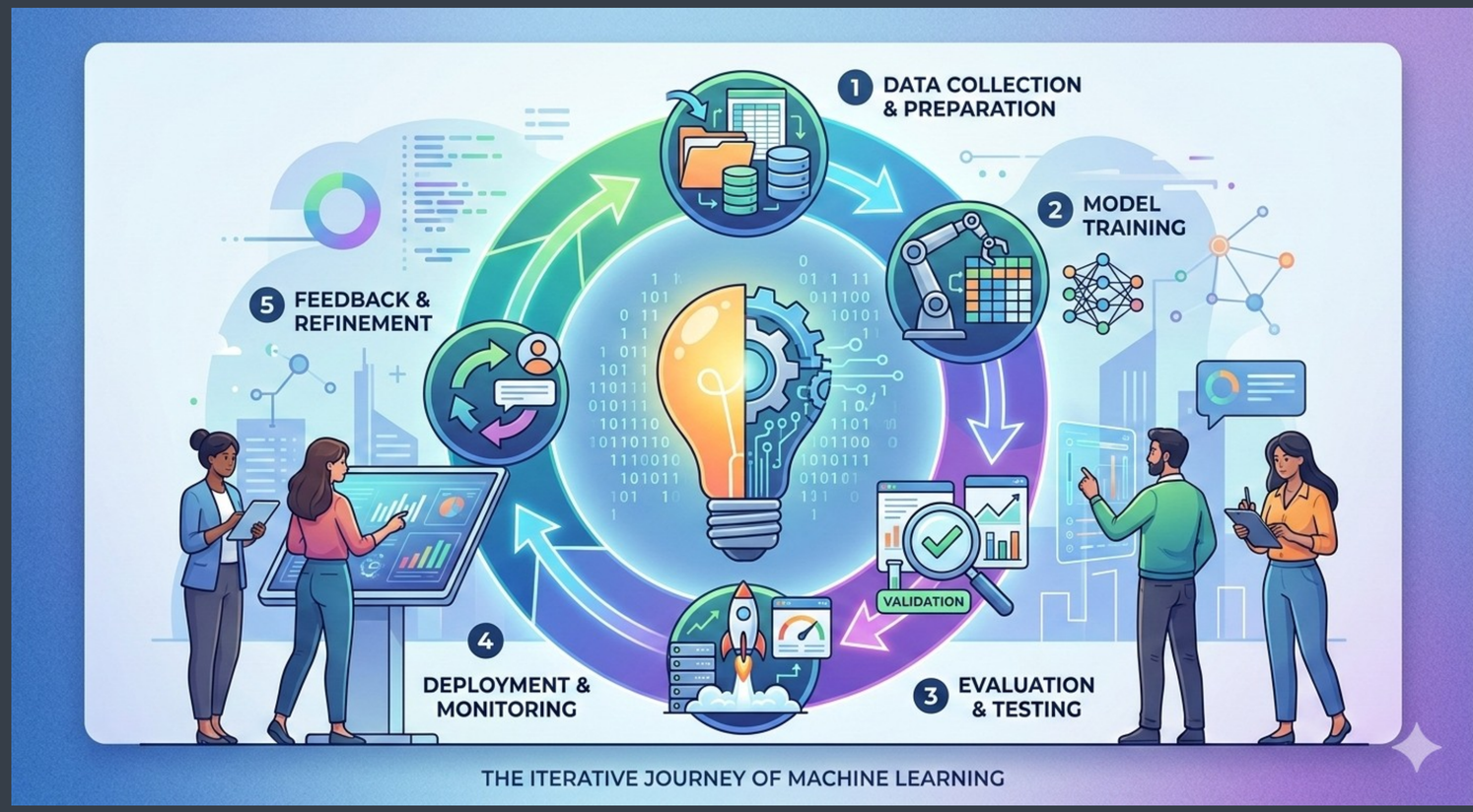


Your New Laboratory

The classroom is now a lab.

Forget passive lectures; expect "Live Code Reviews" and "Decision Defenses."

All your work will live on GitHub, mirroring a professional data science workflow. This course is about establishing a portfolio that proves you can manage the entire machine learning process from start to finish.

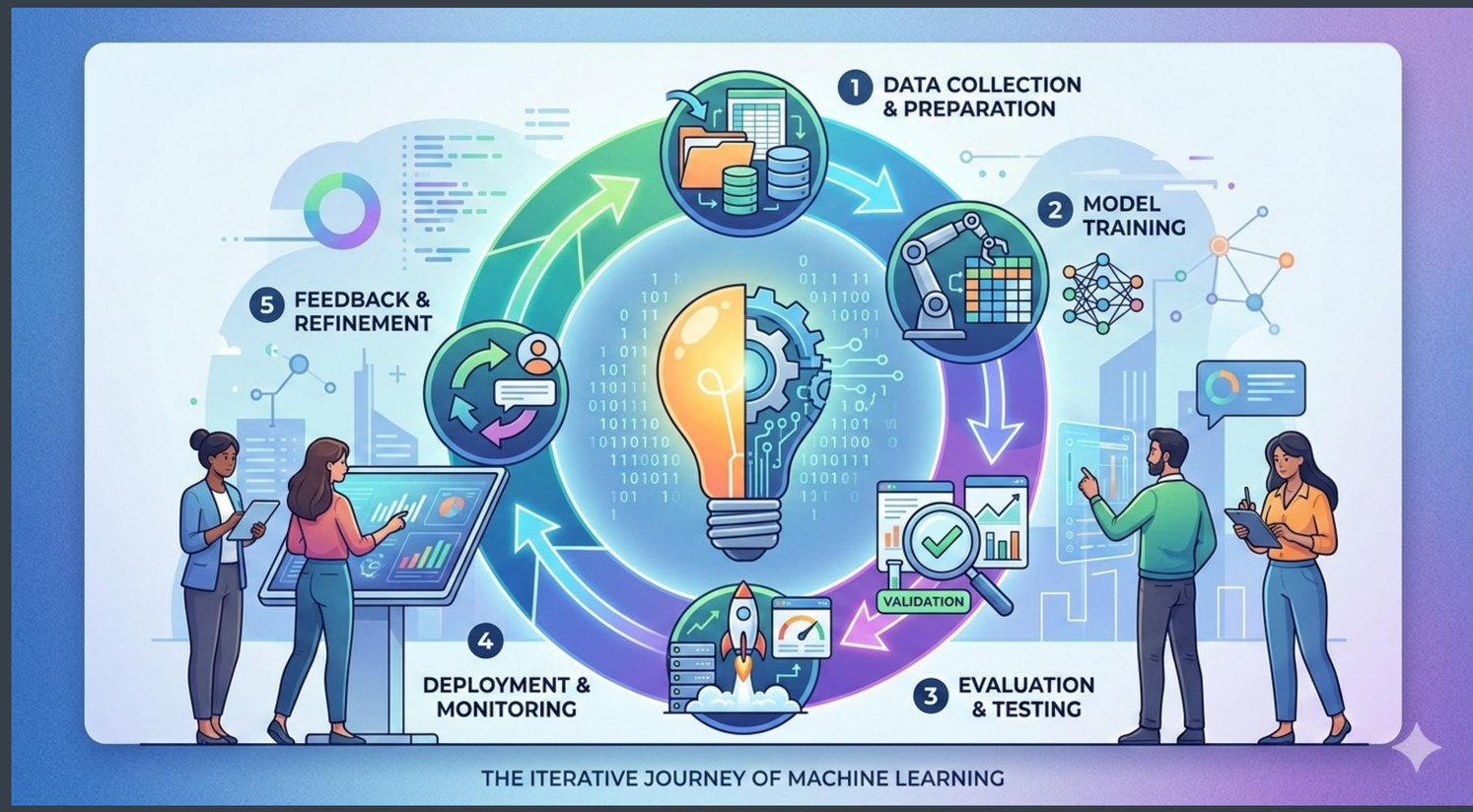


Course Pipeline (The End-to-End View)

We won't march through a textbook of algorithms.

Instead, as your project hits a wall, we will dive into the specific math or logic required to break through.

You will learn KNN, Regression, or Random Forests because your project demands them, making the technical details immediately relevant and memorable.

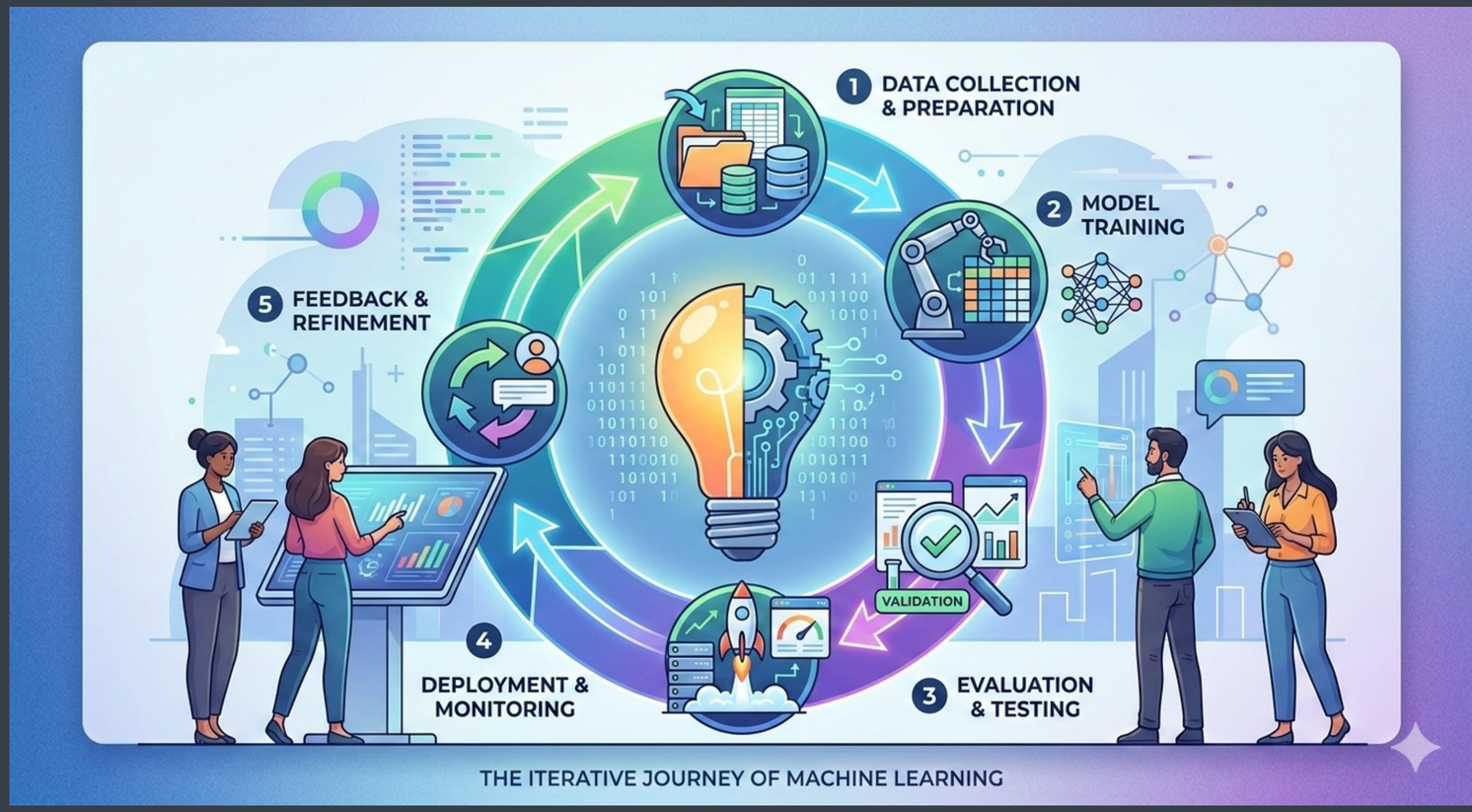


Outcome-Based Grading

Forget traditional midterm anxiety. Your grade in this course is a reflection of your ability to build.

We focus on "cumulative milestones." Each assignment is a building block for your final machine learning project.

You are graded on the integrity of your pipeline and the logic behind your model.

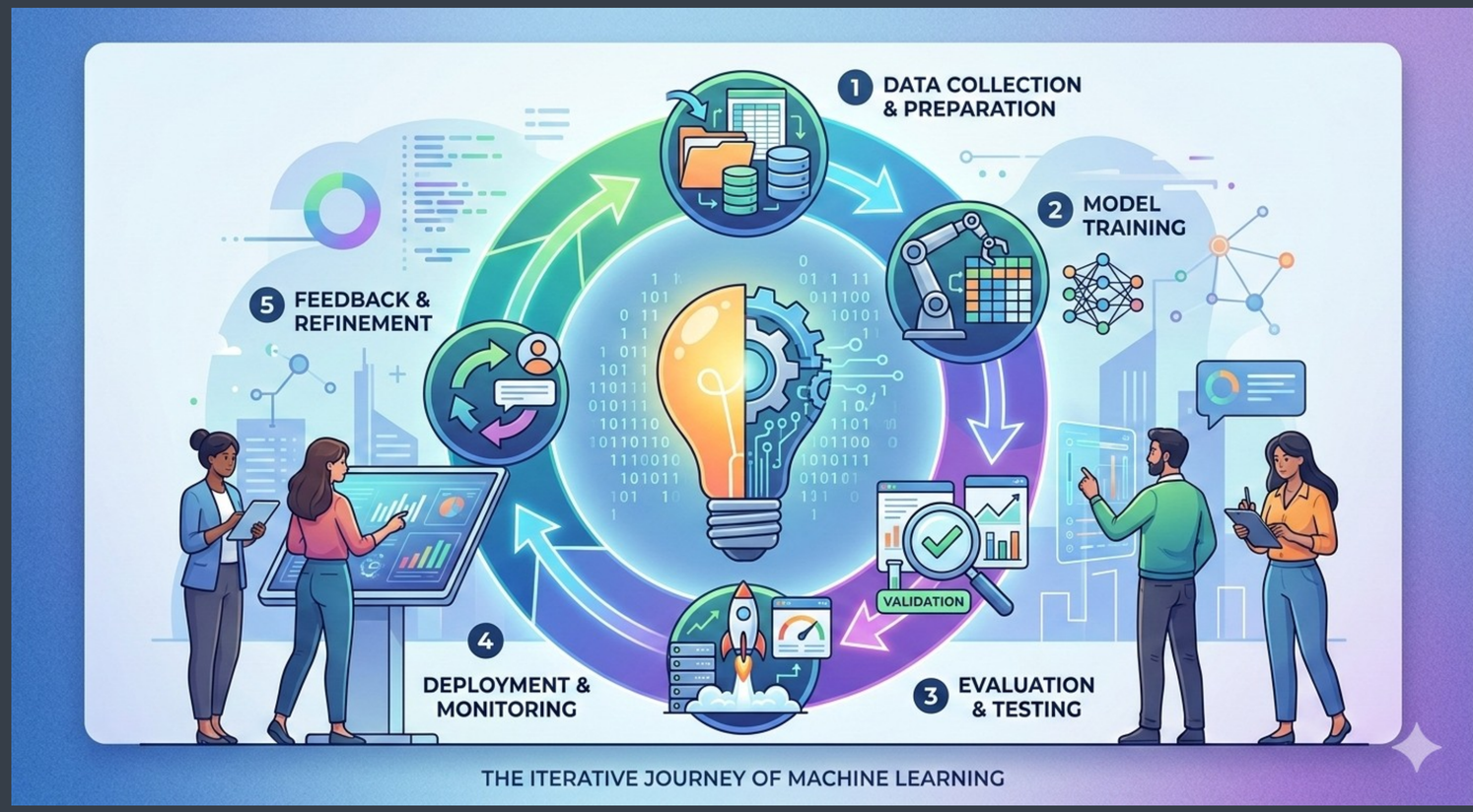


Outcome-Based Grading

Grading follows a professional trajectory.

First, you'll master evaluation metrics by building Confusion Matrices. Then, you'll probably design a basic classifier. These lead to the Final Project, where you solve a real-world economic problem.

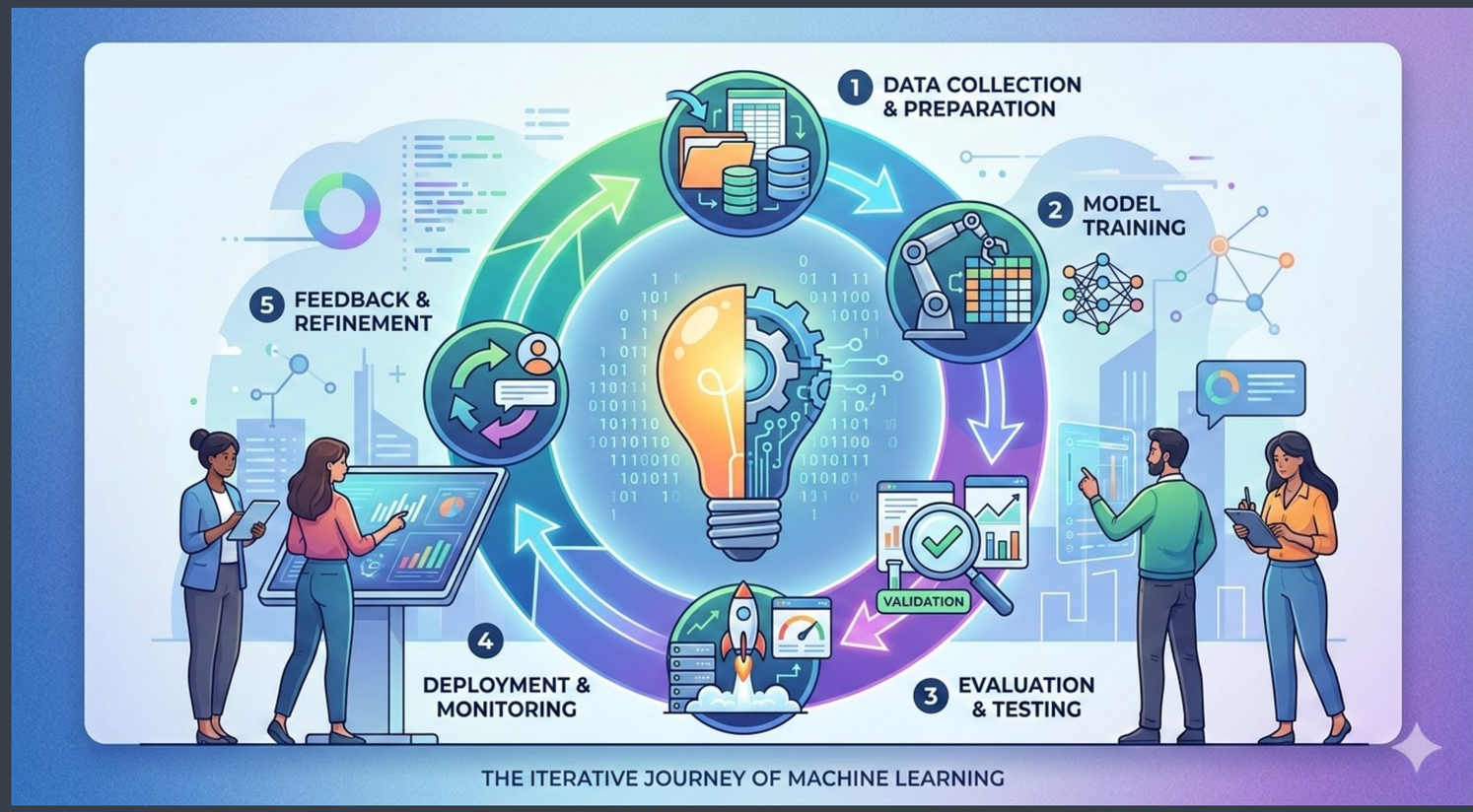
Each step is an opportunity to refine your code based on "Live Code Review" feedback.



Outcome-Based Grading

As we move in steps, **your letter grade will not be effected much by your early mistakes**.

We expect you to make mistakes, and learn along the way.

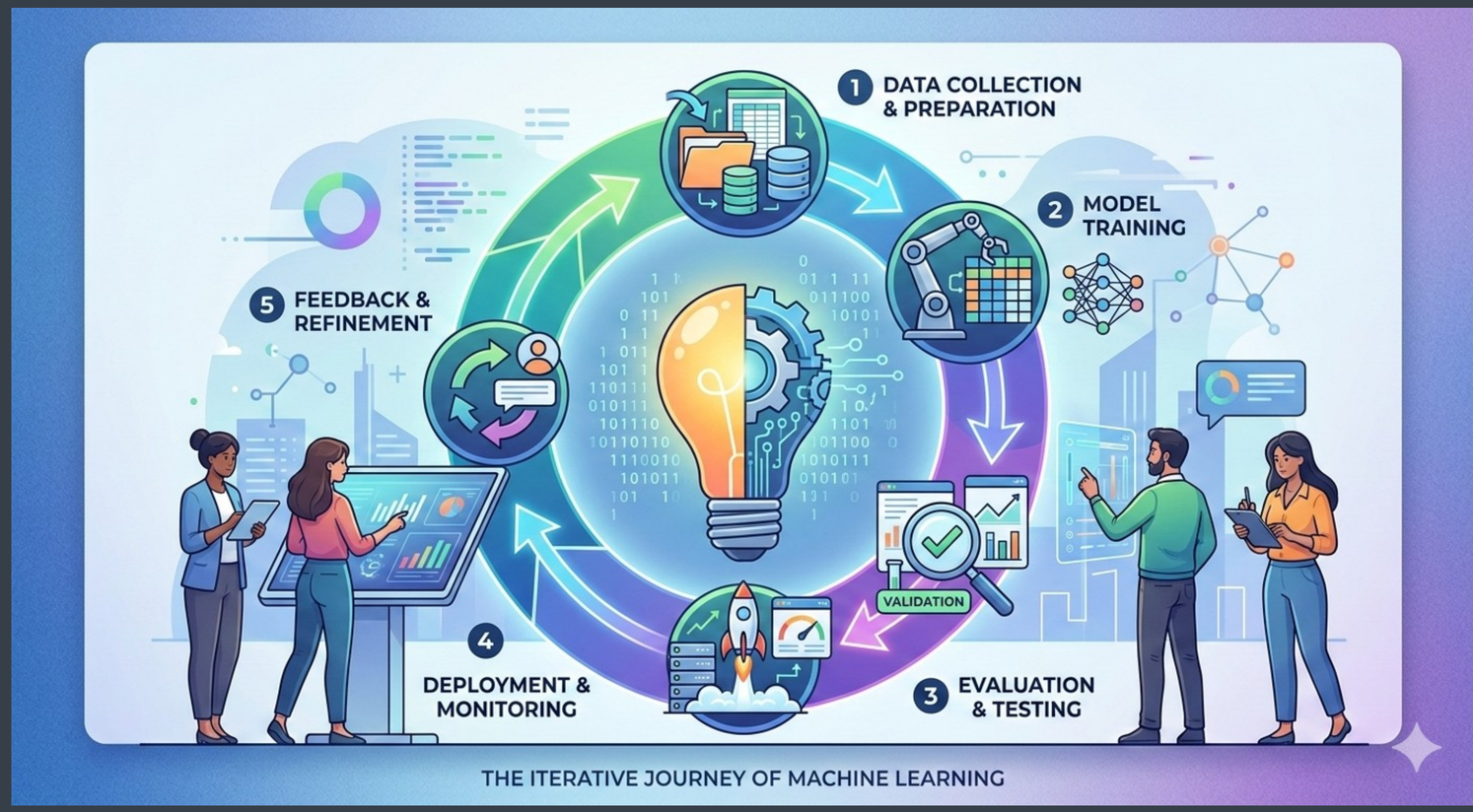


Outcome-Based Grading

The grading structure is designed to be **forgiving but fast-paced**.

Missing early theory won't break your grade, provided you dive into the first coding lab immediately.

Focus on the "First action" items listed on GitHub to align your pace with the rest of the lab.

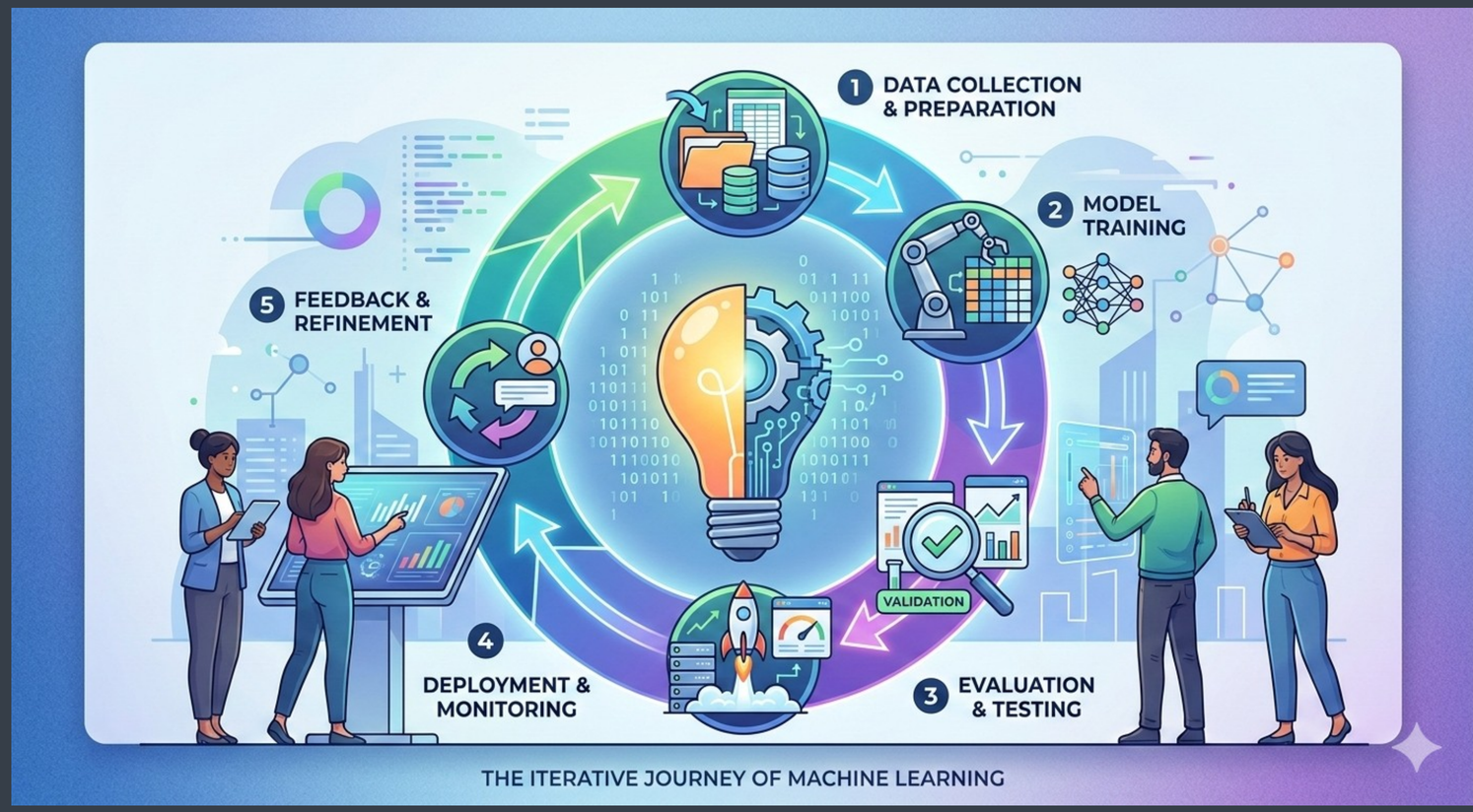


Outcome-Based Grading

On the other hand, a working script isn't enough for an "A."

During "Defense" sessions, you will explain why you chose a specific feature or algorithm.

If an AI helped you code, you must demonstrate you understand the output. We grade your transparency, your troubleshooting process, and your final analytical conclusions.

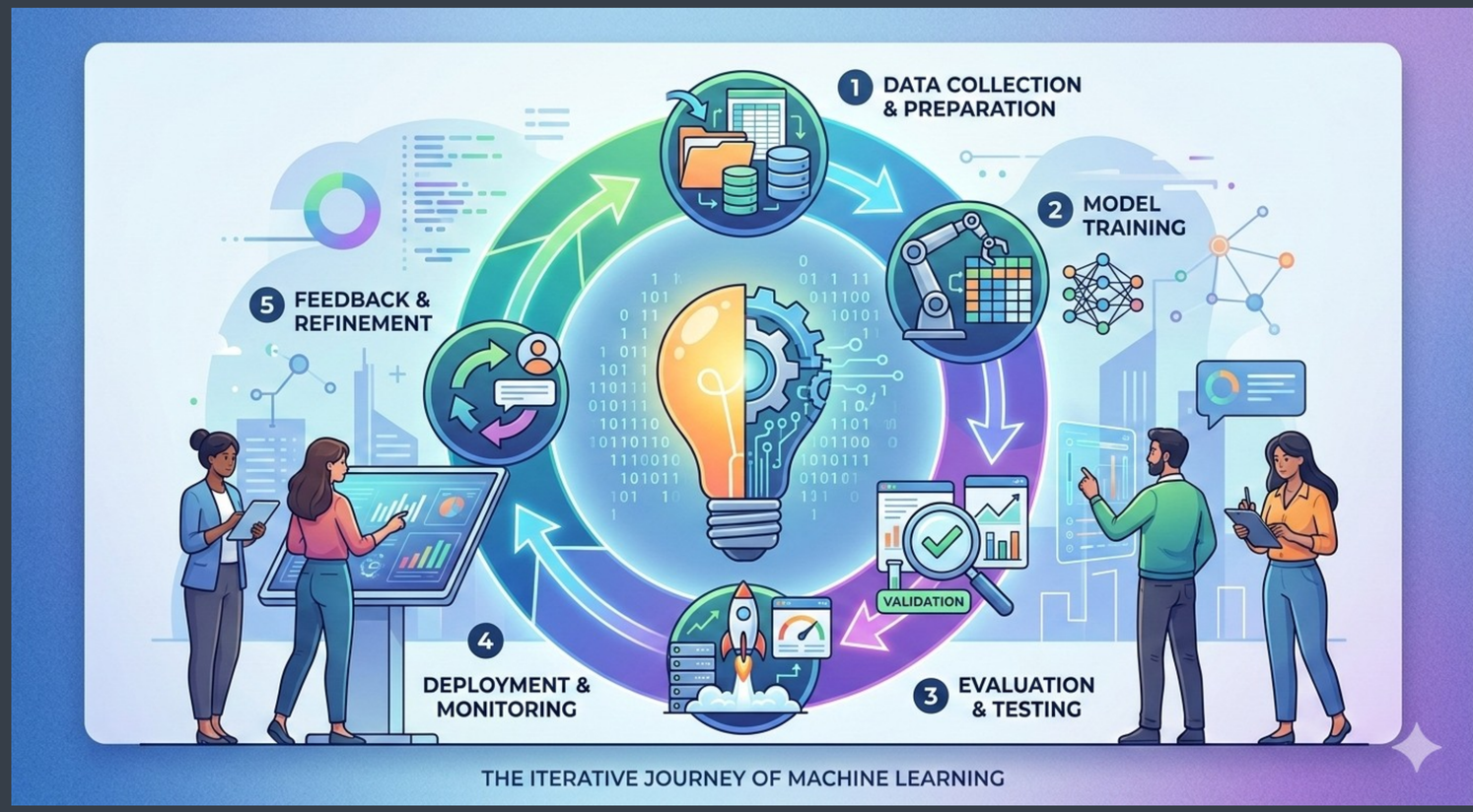


Outcome-Based Grading

All submissions happen via GitHub. Your repository is your resume.

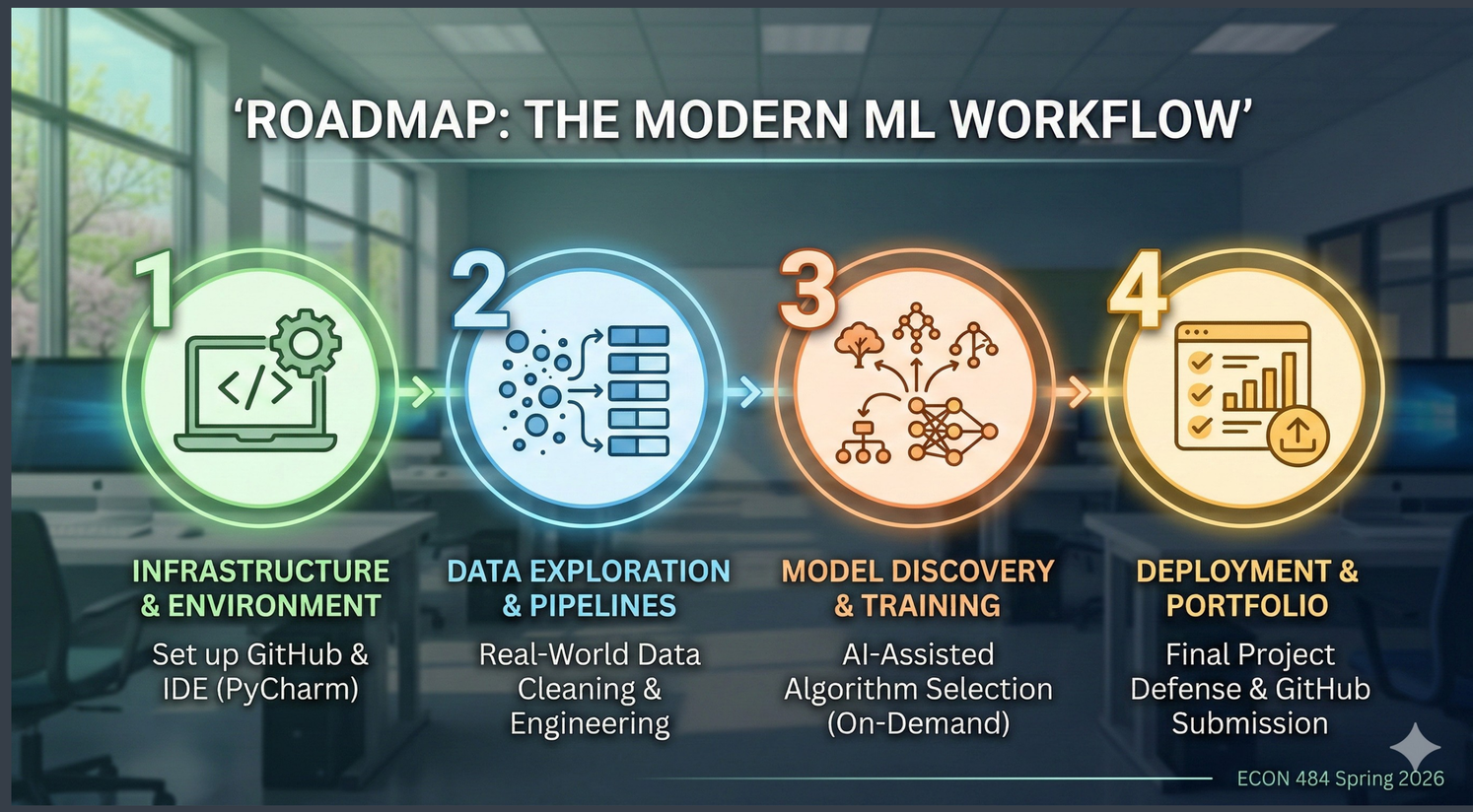
We grade based on version control habits: are your commits meaningful? Is your README clear?

By the end of the semester, you won't just have a grade; you will have a public portfolio showcasing your end-to-end ML capabilities.



The Roadmap

Our journey begins with setting up your infrastructure. In the coming weeks, we will transition from basic classification concepts to building live pipelines. You will move through data exploration, feature selection, and model tuning. Every session is designed to add a specific skill to your professional machine learning toolkit.

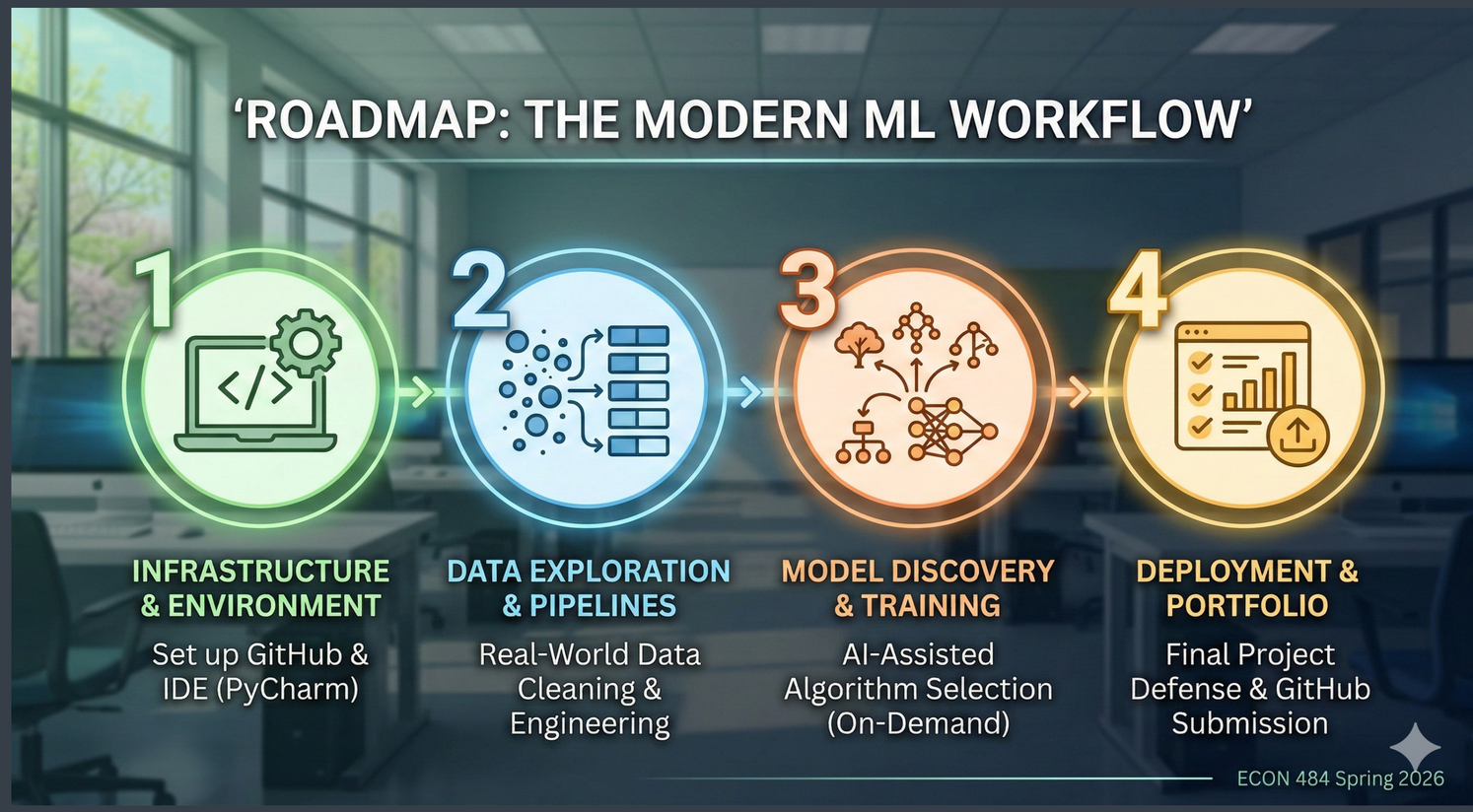


The Roadmap

First, create a GitHub account. GitHub is where your code lives and where I grade your work.

Once registered, complete the "GitHub Foundations" or basic training offered for free on their site.

Mastering version control is non-negotiable for this course and your future career in data science.

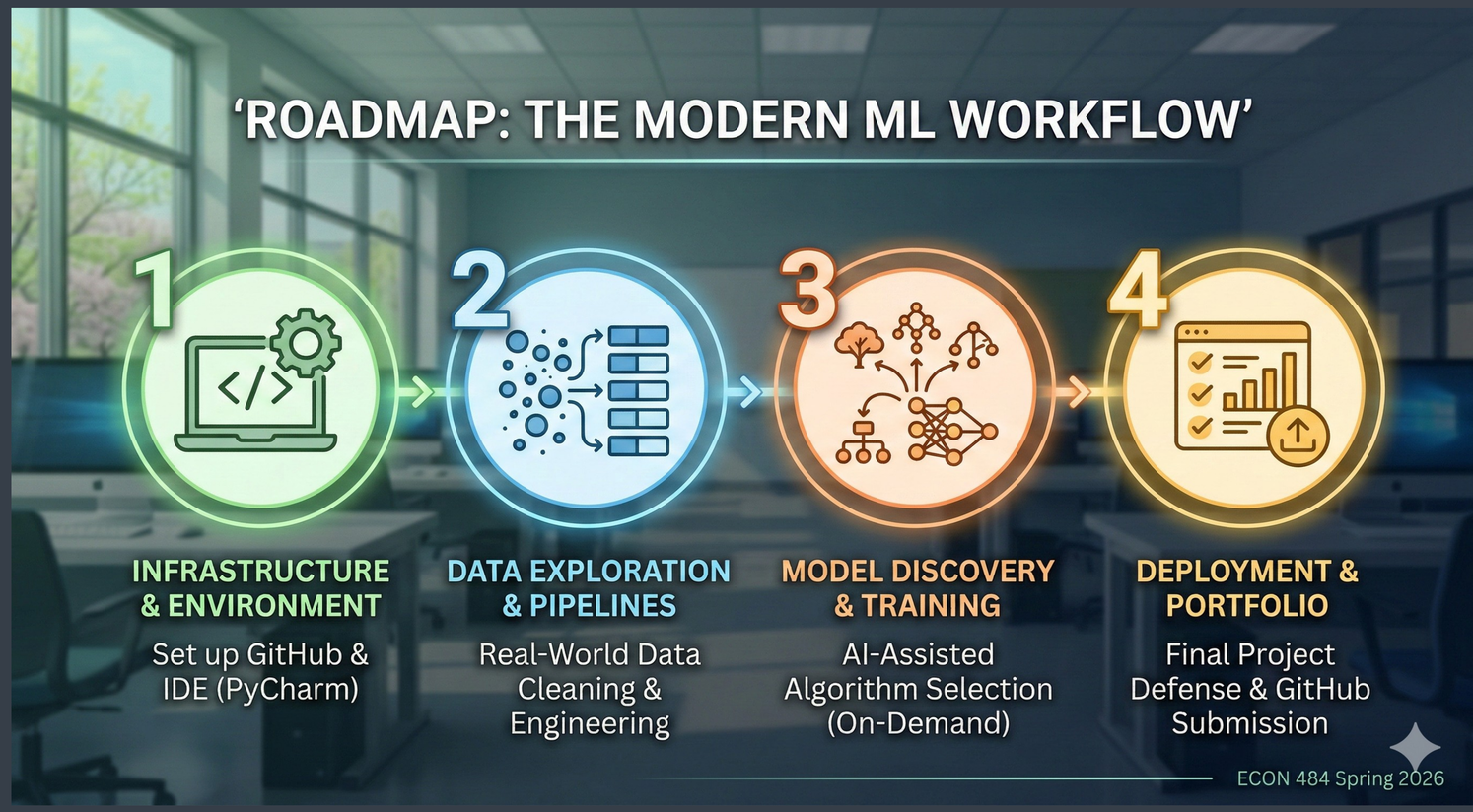


The Roadmap

You need a professional Integrated Development Environment (IDE).

I recommend JetBrains PyCharm. Use your school email address to create a JetBrains account; this grants you a free professional license as a student.

If you prefer VS Code or another tool, ensure it is fully configured for Python development.



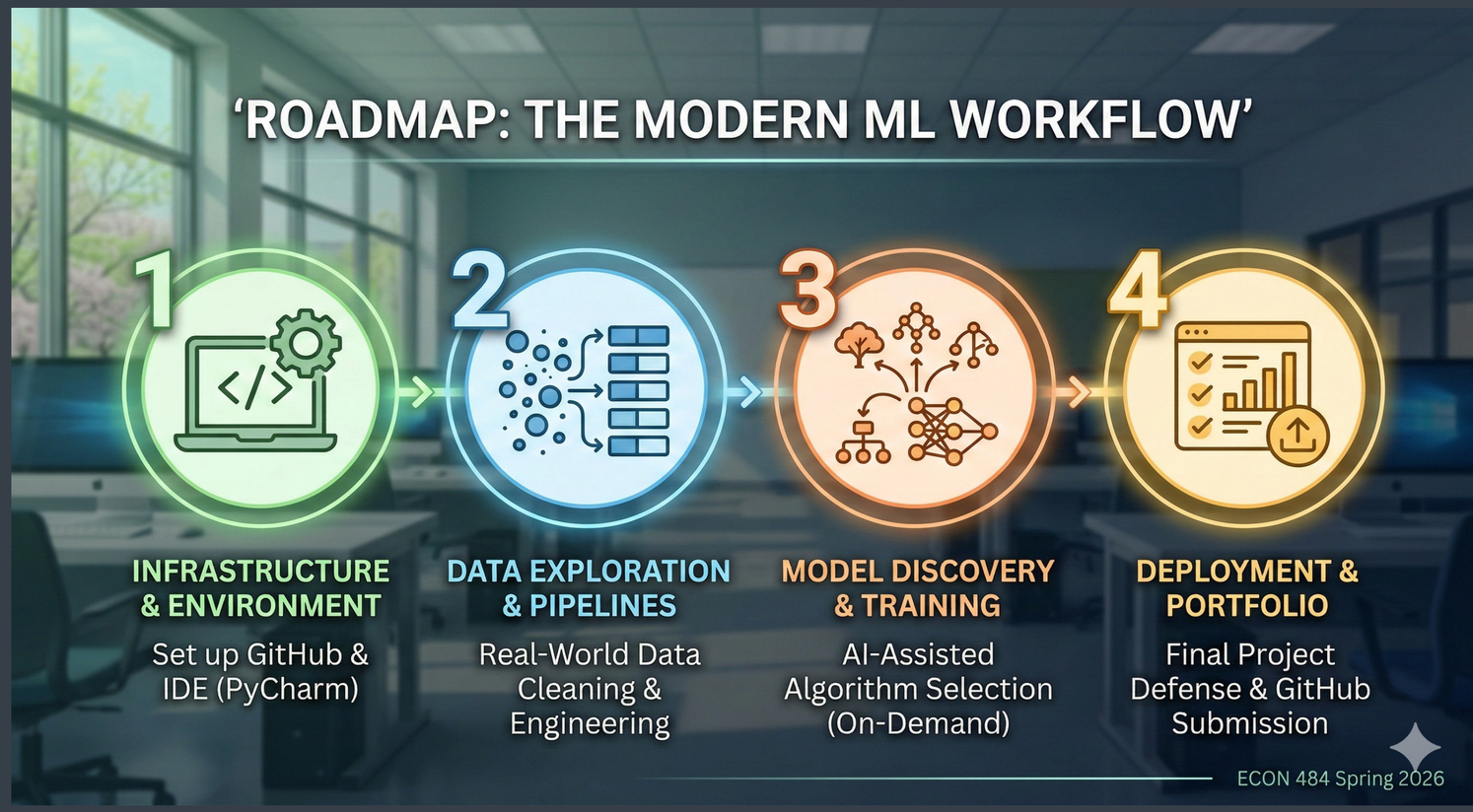
The Roadmap

Start experimenting with LLMs like ChatGPT or Gemini.

Your first task will be to use these tools to automate boilerplate code.

Remember our policy: the AI is your junior developer, but you are the lead architect.

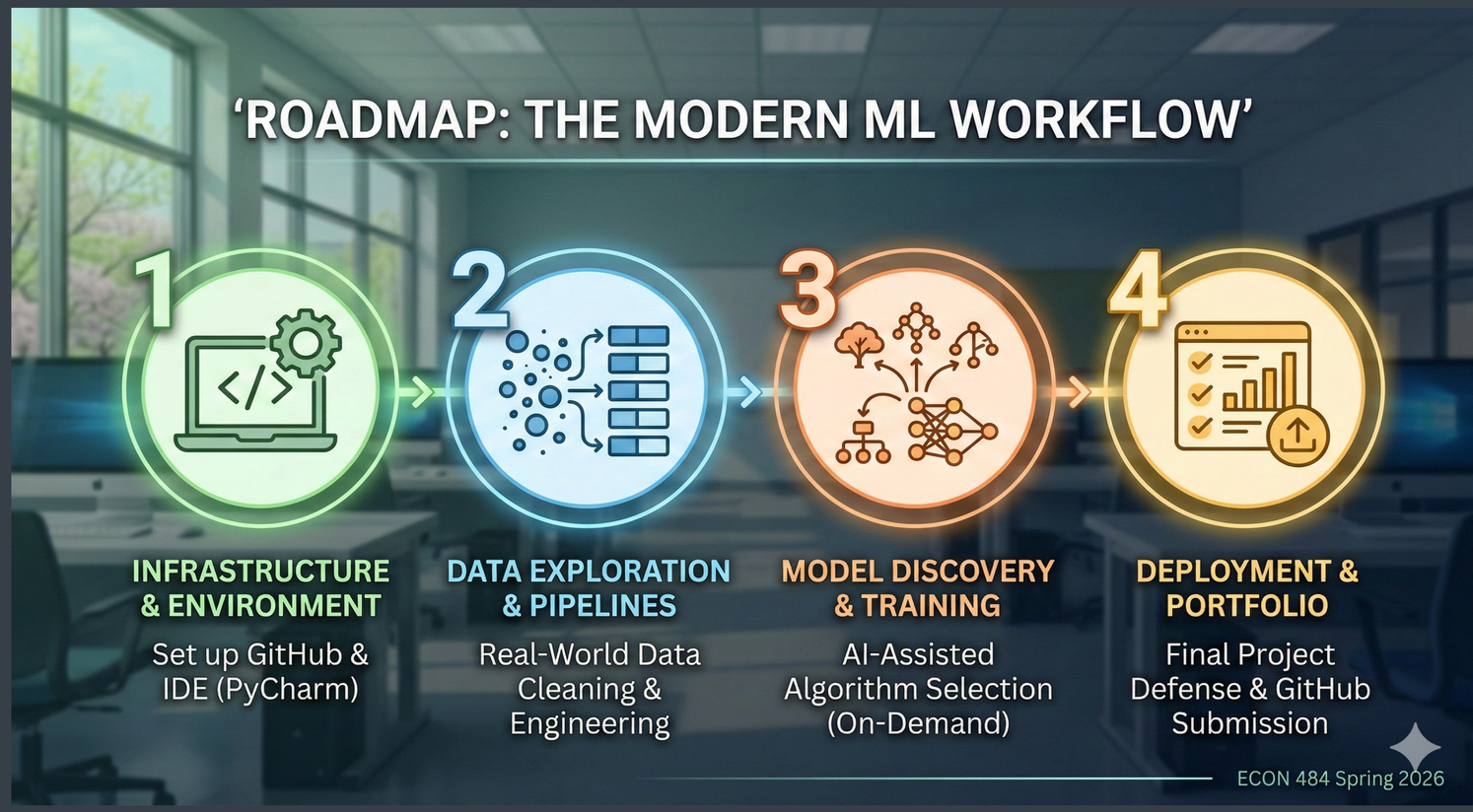
Practice asking the AI to explain the code it generates for you.



The Roadmap

You are now part of a high-intensity data lab.

This course demands curiosity and a "builder" mindset. Check the course repository for the first lab instructions. Let's stop talking about theory and start building the future.



Thank You

Please follow up on Github and your emails (school emails).